Recap

$$\mu_i = \left(\frac{\partial G}{\partial n_i}\right)_{T,P,n_j} = \mu_i^{IM} + \mu_i^{ex} = \underline{G}_i + RT\ln x_i + RT\ln\gamma_i$$

An activity coefficient model relates the activity coefficient of a species in a mixture to the composition of that mixture

Activity coefficient models require data to obtain parameters – this data needs to be measurements of activity coefficient as a function of composition and temperature

Activity coefficients cannot be directly measured, but are inferred from measurements of something at equilibrium with the component of interest – for example we can measure vapour pressure, depression of freezing point or osmotic pressure

Common Activity Coefficient Models
– Random Mixing Models,
– Local Composition Models

Local Composition Models

The local composition around each species is allowed to be different to the average composition

Often a more realistic physical basis

Models remain empirical

Fewer parameters needed and extrapolation better (hopefully!)

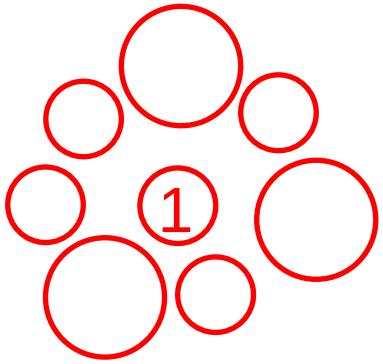
Parameters represent energy difference between interaction of unlike and like molecules – this is inherently binary, so good extrapolation to multicomponent mixtures

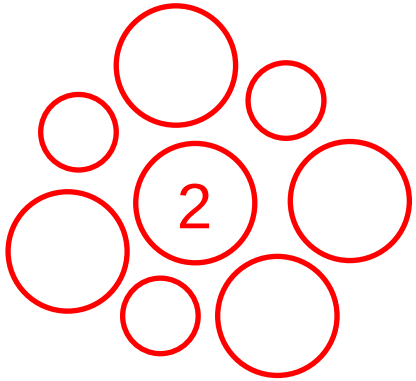
Local compositions are calculated on the basis of a Maxwell-Boltzmann distribution

Wilson equation

1964, the first local composition model

e.g. $x_1 = x_2 = 0.5$





$$x_{11} = \frac{4}{7}$$

$$x_{22} = \frac{4}{7}$$

$$x_{21} = \frac{3}{7}$$

$$x_{12} = \frac{3}{7}$$

$$\frac{G^{ex}}{RT} = -x_1 \ln\left(x_1 + \Lambda_{12} x_2\right) - x_2 \ln\left(x_2 + \Lambda_{21} x_1\right)$$

model parameters
– no more required to use model

interaction energy of 1 and 2

interaction energy of 1 and 1

$$\frac{x_{21}}{x_{11}} = \frac{x_2}{x_1} \exp\left(-\frac{\left(\lambda_{21} - \lambda_{11}\right)}{RT}\right) \qquad \frac{x_{12}}{x_{22}} = \frac{x_1}{x_2} \exp\left(-\frac{\left(\lambda_{12} - \lambda_{22}\right)}{RT}\right)$$

Maxwell-Boltzmann

and $\Lambda_{12} = \frac{V_2}{V_1} \exp\left(-\frac{\left(\lambda_{12} - \lambda_{11}\right)}{RT}\right)$ $\Lambda_{21} = \frac{V_1}{V_2} \exp\left(-\frac{\left(\lambda_{21} - \lambda_{22}\right)}{RT}\right)$

Use $\lambda_{21} - \lambda_{11}$ and $\lambda_{12} - \lambda_{22}$ as parameters – gives best extrapolation

Cannot predict liquid-liquid separation

Non Random Two Liquid Model (NRTL)

An extension of the Wilson equation

One additional parameter – allows liquid-liquid separation

$$\frac{G^{ex}}{RT} = -x_1 x_2 \left[\frac{\tau_{21} G_{21}}{x_1 + x_2 G_{21}} + \frac{\tau_{12} G_{12}}{x_2 + x_1 G_{12}} \right]$$

where $\ln G_{12} = -\alpha \tau_{12}$ $\ln G_{21} = -\alpha \tau_{21}$

and $\frac{x_{21}}{x_{11}} = \frac{x_2}{x_1} \exp\left(-\alpha \frac{(g_{21} - g_{11})}{RT} \right)$ $\frac{x_{12}}{x_{22}} = \frac{x_1}{x_2} \exp\left(-\alpha \frac{(g_{12} - g_{22})}{RT} \right)$

$$\tau_{12} = \frac{(g_{12} - g_{11})}{RT} \qquad \tau_{21} = \frac{(g_{21} - g_{22})}{RT}$$

$g_{21} - g_{11}$ in NRTL equivalent to $\lambda_{21} - \lambda_{11}$ in Wilson

α is new parameter – has some standard settings (see handout)

UNIQUAC (**Uni**versal **Qua**si**c**hemical)

Proposed by Abrams and Prausnitz in 1975

Interaction energies based on surface area

Probably the method of choice in process simulators if other methods are not known to work well

The most flexible method: parameters (hopefully) have small temperature dependence

$$\frac{\underline{G}^{ex}}{RT} = \frac{\underline{G}^{ex}(combinatorial)}{RT} + \frac{\underline{G}^{ex}(residual)}{RT}$$

shape and size

energy

$$\frac{\underline{G}^{ex}(combinatorial)}{RT} = \sum x_i \ln \frac{\phi_i}{x_i} + \frac{z}{2} \sum x_i q_i \ln \frac{\theta_i}{\phi_i}$$

$r_i =$ volume parameter for species i

$q_i =$ surface area parameter for species i

$\theta_i =$ area fraction of species $i = \dfrac{x_i q_i}{\sum x_j q_j}$

$\phi_i =$ volume fraction of species $i = \dfrac{x_i r_i}{\sum x_j r_j}$

$z =$ average coordination number, usually taken to be 10!

Get r_is and q_is by **group contribution** method
– treat as mixture of **functional groups**

$$\frac{\underline{G}^{ex}(residual)}{RT} = \sum_i q_i x_i \ln\left(\sum_j \theta_j \tau_{ji}\right)$$

$$\ln \tau_{ij} = \frac{u_{ij} - u_{jj}}{RT}$$

equivalent of $\lambda_{21} - \lambda_{11}$ in Wilson

2 parameters: $u_{12} - u_{11}$ and $u_{21} - u_{22}$

Fit these parameters to data, as for any other Activity Coefficient model

Group Contribution Methods

Just seen group contribution approach to get combinatorial term in UNIQUAC

Group contribution methods do the same thing for the residual (energy) term – this is a lot less accurate

There are lots of different group contibution methods. One is

UNIFAC (**UNI**QUAC **F**unctional Group **A**ctivity **C**oefficient Model)

UNIFAC (**UNI**QUAC **F**unctional Group **A**ctivity **C**oefficient Model)

This is an extension of UNIQUAC which allows it to be used in situations where data is not available

Parameters evaluated by a group contribution method i.e**. the mixture is envisaged as a mixture of functional groups rather than of molecules**

Parameters for functional groups are tabulated

Details on handout

UNIFAC allows you to produce activity coefficient models without the need for experiment

It is very useful for making quick estimates

But it is also **dangerous** – always **verify** results before using

Summary of Activity Coefficient Models

- Moderately non-ideal systems, all the models perform well

- Regular Solution Theory useful for back of the envelope calcs

- For mixtures of very different species, Wilson, NRTL and UNIQUAC are more reliable. Redlich-Kister with several parameters can also give accurate answers, but the parameters tend to be less reliably extrapolated over temperature

- If liquid phase separation is a possibility, Wilson model is inapplicable. Use Redlich-Kister, NRTL or UNIQUAC

What do you do if experimental data is not available?

1. **Don’t build a chemical plant!**

2. Go and make some VLE binary equilibrium measurements - or better, get someone else to do it: they are painful to do

3. Make a best guess using the UNIFAC method

Advice

Unless there is a good reason not to, use UNIQUAC if you are using a process simulator

If you are doing your own calculations use
the Redlich-Kister expansion

Important points from this section

1. Read it! But no need to memorise
2. You should be able to use Margules, RST and van Laar equations for hand calculations, but you don't need to remember them (except one constant Margules)
3. Can get multi-component (>2) models by combining parameters for binary mixtures for all models
4. Hence activity coefficient models allow you to extrapolate over temperature, composition and to multi-component mixtures
5. Activity coefficients are just **fits** of functions to experimental data. It is therefore dangerous to extrapolate beyond the range of temperature and composition over which the data was collected
6. In assessing the reliability of models, you need to know about the model, but also about the data from which the parameters were calculated
7. There is no point in using a very sophisticated model, with parameters based on limited data